
SAARLAND UNIVERSITY

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Machine Learning for Natural Language Processing
Term Paper



PROTEIN LANGUAGE MODEL ON LONG SEQUENCE GENERATION

submitted by
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Abstract

Protein language models (PLMs) have achieved remarkable progress in both discriminative and generative protein modeling tasks, particularly under biologically meaningful constraints such as InterPro (IPR) domains, Gene Ontology terms, and enzyme classifications. Recent advances, including multi-constraint diffusion-based frameworks such as CFP-GEN, demonstrate state-of-the-art performance on standard benchmarks. However, the ability of such models to generalize to long protein sequence generation ($>1,000$ amino acids) remains largely unexplored, despite the biological and clinical relevance of large multi-domain proteins.

In this work, I systematically evaluate CFP-GEN on long-sequence conditional generation using a curated Swiss-Prot dataset of proteins between 1,000 and 2,000 residues. I compare its performance against its unconditional backbone, DPLM, under controlled IPR constraints. Evaluation using Mean Reciprocal Rank (MRR) and Maximum Mean Discrepancy (MMD) reveals that while CFP-GEN maintains comparable ranking performance, it does not outperform—and in some cases underperforms—the unconditional baseline in distributional alignment for long sequences. Performance degradation increases with sequence length for both models, with multi-constraint conditioning exhibiting reduced generalization in extended contexts.

My findings suggest that incorporating biological constraints alone does not resolve long-context generation challenges in PLMs. I highlight the need for architectural and training innovations specifically targeted at long-sequence protein modeling to bridge this gap.

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Chapter 1

Introduction

1.0.1 Background

Protein language models (PLMs) have recently demonstrated substantial impact across a wide range of biotechnological applications, including drug discovery, enzyme engineering, and de novo therapeutic protein design [12, 5, 2, 11, 8]. By learning statistical regularities from large-scale protein sequence corpora, these models enable the generation of novel proteins that may not exist in nature or would otherwise require extensive evolutionary timescales to emerge [1, 4, 9].

Such capabilities are particularly relevant for addressing complex diseases, including cancer and developmental syndromes, where large, multi-domain proteins often play central regulatory roles. However, many clinically significant proteins exceed 1,000 amino acids in length. Consequently, to fully realize their translational potential, PLMs must reliably generate accurate long protein sequences (>1,000 residues), preserving global structural coherence and functional constraints across extended contexts.

1.0.2 Motivation

Despite their success, PLMs inherit fundamental limitations from large language models in natural language processing, particularly in long-sequence generation. These challenges arise from restricted context windows, quadratic attention complexity, optimization instability, and limited long-sequence representation in training corpora.

Protein sequences introduce additional complexity. Unlike natural language, proteins are composed of only 20 amino acids, meaning that small perturbations can significantly affect folding, domain organization, and function. Therefore, accurate generation requires precise modeling of long-range dependencies and structural compatibility.

In contrast to unconditional language models, protein generation often requires satisfying multiple biological constraints simultaneously, such as InterPro (IPR) domains, Gene Ontology (GO) annotations, or enzyme classifications. To address this, prior PLMs have adopted diverse training paradigms—including masked language modeling, au-

toregressive generation, and diffusion-based objectives—across sequence and structure modalities.

While these approaches demonstrate strong performance across discriminative and generative benchmarks, existing methods do not systematically address the fundamental limitation of long-sequence generation under biologically meaningful constraints. In particular, it remains unclear whether multi-constraint conditioning improves or hinders generation quality for sequences beyond typical training lengths.

1.0.3 Contribution

In this work, I investigate whether current state-of-the-art PLMs—specifically CFP-GEN—effectively generalize to long protein sequence generation (>1,000 residues) under conditional settings.

My contributions are as follows:

- I construct a curated long-sequence evaluation dataset (1,000–2,000 amino acids) derived from Swiss-Prot, ensuring removal of training overlap and controlling for unseen IPR conditions.
- I systematically compare CFP-GEN against its unconditional backbone model DPLM.
- I analyze performance trends across increasing sequence lengths to assess generalization under extended contexts.
- I provide an open-source experimental framework to facilitate future research on long-sequence protein modeling.¹

¹<https://github.com/KhondokerIslam/ml4lp>

Chapter 2

Related Work

2.0.1 Existing Methods

PLMs have been extensively applied to various protein science tasks, which include discriminative and generative tasks. ESM2 [7] is the pioneer model to explore this direction by uncovering patterns of protein sequence through masked language modeling objective on a number of discriminative tasks. Subsequently, ProtGPT2 [3] performed unconditional protein generation similar to natural language generation. Subsequent masked learning objective models, such as ProGen [10] and ProteoGAN [6], onboarded several features, such as protein family tags and GO terms, to generate informed generation. Later, EvoDiff [1] introduces a discrete diffusion model instead of the masked language modeling objective. This enables conditional protein generation, such as sequence inpainting and MSA-guided generation.

Consequently, DPLM [11] is proposed as a new foundation model that not only supports representation learning for discriminative tasks but also enables conditional generation guided by secondary structure classifications and sequence, allowing better knowledge on protein understanding at the atomic level. Despite these advances, existing models primarily incorporate single or limited conditioning signals. Systematic integration of multiple biological constraints within a unified framework remains underexplored, even though multi-condition incorporation is intuitively important for capturing the complex, hierarchical dependencies that govern protein structure and function at the atomic level.

2.0.2 CFP-GEN

Recently, CFP-GEN proposed a multi-constraint conditional protein generation framework built upon DPLM. The model introduces four training frameworks designed to disentangle and integrate constraint embeddings effectively. These include co-design strategies for sequence inpainting and constraint embedding parameter updates, enabling improved attribution of generated sequence features to specific conditioning signals.

Although trained predominantly on small-to-medium sequence lengths, CFP-GEN demonstrates state-of-the-art performance across multiple discriminative and generative benchmarks. Its multi-constraint design represents a promising direction for structured protein generation under biological priors.

2.0.3 Goal

This work investigates whether CFP-GEN generalizes effectively to long-sequence protein generation (1,000–2,000 residues) under a single constraint setting (IPR). Given that the original study reports limited coverage of extensive constraint combinations, we additionally explore whether increased constraint exposure during training impacts long-sequence performance.

My **research question** is as follows: *Does multi-constraint conditioning improve generalization for long protein generation, or does it exacerbate long-context modeling limitations inherent to PLMs?*

Chapter 3

Experiment

This section describes the dataset construction, experimental setup, and evaluation methodology designed to assess long-sequence protein generation.

3.0.1 Dataset

Data Collection

I curated long protein sequences from UniProt Swiss-Prot² by filtering entries with sequence lengths exceeding 800 residues. Approximately 37,000 sequences containing annotated InterPro (IPR) domains, amino acid sequences, and entry identifiers were initially collected. I subsequently applied the following filtering criteria:

- Retained sequences between 1,000 and 2,000 amino acids.
- Removed sequences overlapping with the CFP-GEN training set.
- Excluded samples whose IPR annotations were entirely unseen during training to avoid evaluating unconditional generalization.
- This procedure ensures a controlled evaluation of conditional long-sequence generation.

Data Statistics

Figure 3.1 illustrates the sequence length distribution. Notice that a clear imbalance is observed, with sequences near 1,000 residues substantially more frequent than those approaching 2,000 residues. As CFP-GEN model lacked substantial IPR count during training, Figure 3.2 presents the distribution of unseen IPR annotations. The harmonic mean between 6 and 10 missing IPRs suggests moderate constraint sparsity.

²<https://www.uniprot.org/>

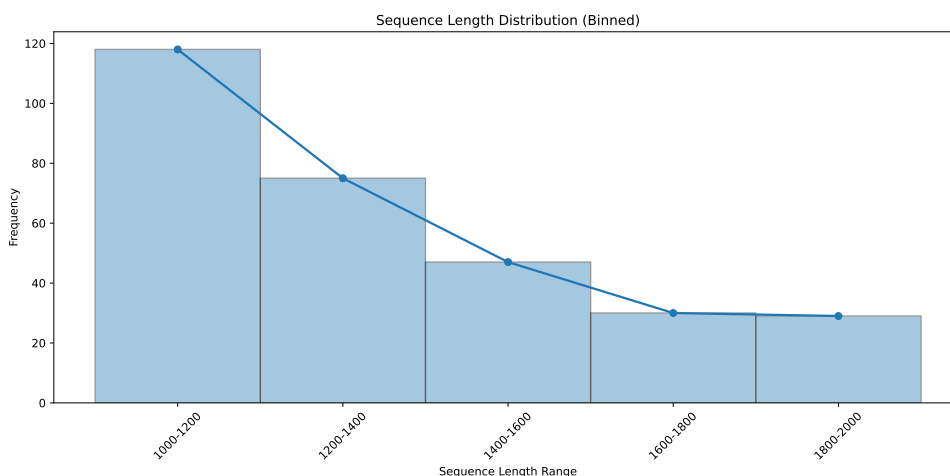


Figure 3.1: Frequency length distribution of the dataset.

To assess whether missing IPRs correlate with sequence length, Figure 3.3 provides a box-plot of unseen IPR counts across length bins. The mean remains consistently between 6 and 10 across bins, with higher variance observed between 1,400–1,800 residues. This analysis indicates that performance degradation across length ranges cannot be attributed solely to missing constraint counts.

3.0.2 Methodology

Implementation

I conduct my experiment on a pretrained CFP-GEN 650m parameter model. This model was trained on 103,939 protein sequences from Swiss-Prot annotated with 1,154 InterPro (IPR) domains and respective Gene Ontology and Enzyme datasets. During evaluation, I specified the models to generate sequences between 1k and 2k and kept the sample strategy as it is, i.e., `gumbel_argmax`. To understand the model’s performance, I chose the unconditional pretrained DPLM as the baseline.

Evaluation Metric

To evaluate the model’s performance, I adopt Mean Reciprocal Rank (MRR) and Maximum Mean Discrepancy (MMD), which assess the similarity between the generated and the real sequences. Moreover, both metrics are consistent with the evaluation protocol of CFP-GEN. Important to note, the higher the MRR, the better the performance, and vice versa for MMD.

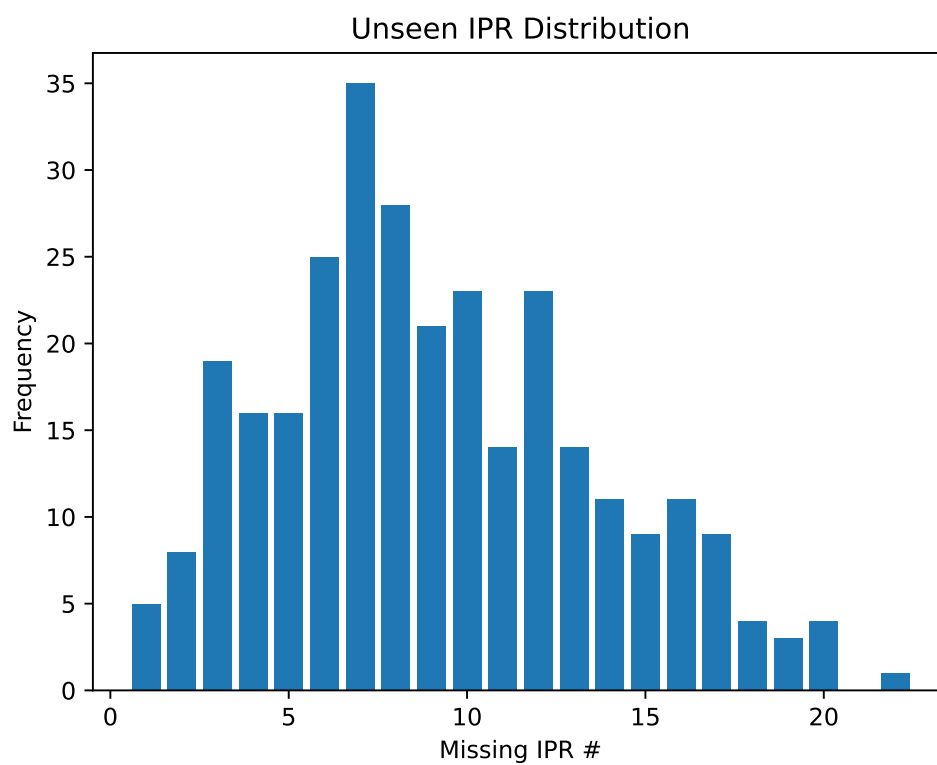


Figure 3.2: Unseen IPR distribution of the dataset.

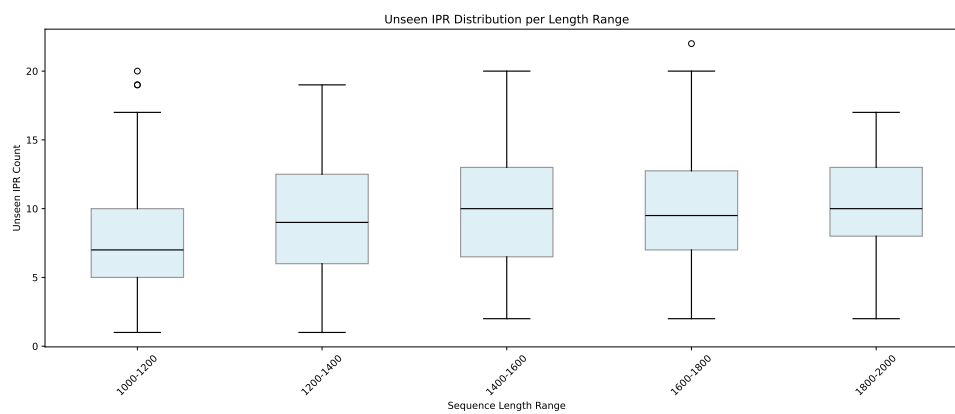


Figure 3.3: Unseen IPR Distribution per Length Range.

Chapter 4

Findings

4.0.1 Results

Table 4.1 compares CFP-GEN and DPLM on long-sequence generation. While CFP-GEN achieves comparable MRR to the baseline, the unconditional DPLM demonstrates superior performance on MMD. This suggests that multi-constraint conditioning does not provide measurable benefit for long-sequence generation and may negatively impact distributional fidelity. These results indicate that constraint embedding updates in CFP-GEN may not effectively propagate across extended sequence contexts.

Model	MRR $\uparrow$	MMD $\downarrow$
DPLM	0.007	0.324
CFP-GEN	0.007	0.466

Table 4.1: MRR and MMD for the baseline and CFP-GEN model.

4.0.2 Discussion

Figure 4.1 visualizes MMD trends across increasing sequence lengths. Both models exhibit approximately linear degradation as sequence length increases, reflecting long-context limitations inherent to PLMs. However, CFP-GEN consistently underperforms the unconditional baseline, suggesting that multi-constraint conditioning may exacerbate long-range modeling challenges.

Interestingly, while DPLM shows negligible degradation at the highest length bin, CFP-GEN exhibits accelerated performance decline beyond linear trends. This behavior indicates limited generalization capacity when constraint embeddings must interact over extended contexts. Future work should investigate architectural modifications beyond constraint incorporation—such as hierarchical modeling, segment-wise diffusion, or memory-efficient attention mechanisms—to address long-sequence degradation in PLMs.

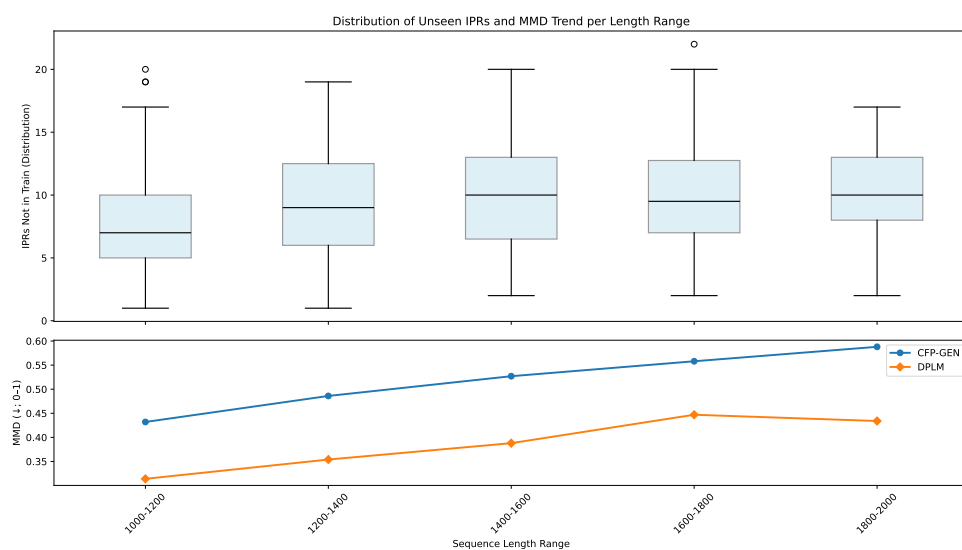


Figure 4.1: Scatter plot distribution of CFP-GEN and DPLM across length size.

Chapter 5

Conclusion

This study investigates whether multi-constraint protein language models generalize effectively to long-sequence protein generation. By evaluating CFP-GEN on sequences between 1,000 and 2,000 amino acids and comparing it against its unconditional backbone DPLM, I provide a focused analysis of long-context conditional generation.

My results indicate that although CFP-GEN retains comparable performance to the baseline in ranking-based evaluation (MRR), it does not demonstrate improved distributional fidelity (MMD) for long sequences. Moreover, both models exhibit performance degradation as sequence length increases, highlighting persistent long-context limitations in current PLMs. Notably, multi-constraint conditioning does not appear to mitigate this degradation and may introduce additional optimization challenges in extended sequence regimes.

These findings suggest that constraint incorporation alone is insufficient to address the long-sequence generation problem. Instead, future research should explore architectural modifications, hierarchical modeling strategies, and length-aware training objectives specifically designed for extended protein contexts. Addressing this limitation is critical for enabling reliable generation of large, multi-domain proteins that are central to many disease pathways and therapeutic applications.

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